Data set $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{N}$ ← evidence

N: number of data points.

$x_i \in \mathbb{R}^n$ (inputs / features / attributes)

$y_i \in \mathbb{R}$ (labels / outputs)

NOT KNOWN: functional relationship f s.t.

$$y = f(x)$$

Choose a set of functions (basis functions)

$$B = \{f_1, f_2, \ldots, f_n\}$$

Approximate the function relationship f as a linear combination of these basis functions.

$$f(x) \approx \alpha_1 f_1(x) + \alpha_2 f_2(x) + \cdots + \alpha_k f_k(x)$$

Find α_i's, $\alpha = \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_k \end{bmatrix} \in \mathbb{R}^k$ such that this linear combination approximates $f(x)$ in the best possible way.

For fixed α_i's and ($x_1, \cdots, x_N$ given data)

$$r(x_i) = y_i - (\alpha_1 f_1(x_i) + \alpha_2 f_2(x_i) + \cdots + \alpha_k f_k(x_i))$$

$r(x_i)$: residual

y_i: observed label

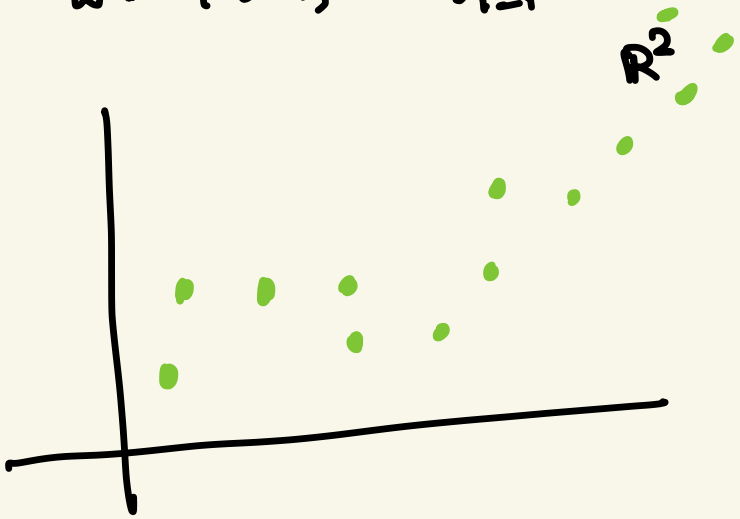
$(\alpha_1 f_1(x_i) + \alpha_2 f_2(x_i) + \cdots + \alpha_k f_k(x_i))$: prediction

for every $i = 1, 2, \cdots, N$

Typical example:

$$\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{N}$$

$x_i \in \mathbb{R}$, $y_i \in \mathbb{R}$



$$\begin{cases} f_1(x) = 1 \\ f_2(x) = x \\ \vdots \\ f_k(x) = x^{k-1} \end{cases}$$

basis functions

$y = f(x)$ ← unknown

$$f(x) \approx \underbrace{\alpha_1 + \alpha_2 x + \alpha_3 x^2 + \cdots + \alpha_k x^{k-1}}$$

Best approximation. (loss function)

$$\min_{\alpha \in \mathbb{R}^k} \sum_{i=1}^{N} |r_i| \qquad \min_{\alpha \in \mathbb{R}^k} \sum_{i=1}^{N} r_i^2$$

The parameter vector $\hat{\alpha} \in \mathbb{R}^k$, $(\hat{\alpha}_1, \hat{\alpha}_2, \cdots, \hat{\alpha}_k)$

for which $\sum_{i=1}^{N} r_i^2$ is minimum.

$$\hat{\alpha} = \underset{\alpha \in \mathbb{R}^k}{\operatorname{argmin}} \sum_{i=1}^{N} r_i^2$$

minimum value of the function. : min

minimizer : argmin

min $f(x)$

$f(x) = x^2 + 1$

min $f(x) = 1$

$$\hat{y}_i = \alpha_1 f_1(x_i) + \alpha_2 f_2(x_i) + \cdots + \alpha_k f_k(x_i)$$

for $i = 1, 2, \ldots, N$

$$= \underbrace{\begin{bmatrix} f_1(x_i) & f_2(x_i) & \cdots & f_k(x_i) \end{bmatrix}}_{k\text{-vector}} \underbrace{\begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_k \end{bmatrix}}_{k\text{-vector}}$$

$$\hat{y} = \begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \vdots \\ \hat{y}_i \\ \vdots \\ \hat{y}_N \end{bmatrix} = \overbrace{\begin{bmatrix} f_1(x_1) & f_2(x_1) & \cdots & f_k(x_1) \\ f_1(x_2) & f_2(x_2) & \cdots & f_k(x_2) \\ \vdots & & & \\ f_1(x_i) & f_2(x_i) & \cdots & f_k(x_i) \\ \vdots & & & \\ f_1(x_N) & f_2(x_N) & \cdots & f_k(x_N) \end{bmatrix}}^{\Phi \in \mathbb{R}^{N \times k}} \underbrace{\begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_k \end{bmatrix}}_{k \times 1}_{= \alpha}$$

$$\boxed{\hat{y} = \Phi \alpha}$$

Matrix representation

In the case of basis $\{1, x, x^2, \dots, x^{k-1}\}$

$$\Phi = \begin{bmatrix} 1 & x_1 & x_1^2 & \cdots & x_1^{k-1} \\ 1 & x_2 & x_2^2 & \cdots & x_2^{k-1} \\ \vdots & & & & \\ 1 & x_N & x_N^2 & \cdots & x_N^{k-1} \end{bmatrix} \in \mathbb{R}^{N \times k}$$

Vandermonde matrix.

Residuals

$$r = \begin{bmatrix} r_1 \\ r_2 \\ \vdots \\ r_N \end{bmatrix} = \begin{bmatrix} y_1 - \hat{y}_1 \\ y_2 - \hat{y}_2 \\ \vdots \\ y_N - \hat{y}_N \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_N \end{bmatrix} - \hat{y} = y - \hat{y}$$

$$= y - \Phi\alpha$$

$$\sum_{i=1}^{N} r_i^2 = [r_1 \; r_2 \; \cdots \; r_N] \begin{bmatrix} r_1 \\ r_2 \\ \vdots \\ r_N \end{bmatrix}$$

$$= r^T r$$

$$= (y - \Phi\alpha)^T (y - \Phi\alpha)$$

$$= \|y - \Phi\alpha\|_2^2 \quad \leftarrow \text{spectral norm or 2-norm of a vector.}$$

Loss fn minimization:

$$\min_{\alpha \in \mathbb{R}^k} \sum_{i=1}^{N} r_i^2 = \min_{\alpha \in \mathbb{R}^k} \|y - \Phi\alpha\|_2^2$$

$$\min_{\alpha \in \mathbb{R}^k} \| y - \Phi\alpha \|_2^2$$

→ Possible minimum value can be zero.

i.e. $\exists\ \hat{\alpha} \in \mathbb{R}^k$ s.t. $y - \Phi\hat{\alpha} = 0 \in$ zero vector.

$\Rightarrow \| y - \Phi\hat{\alpha} \|_2^2 = 0 \in$ real no. zero.

$$\Phi\hat{\alpha} = y$$

(The system of eqns $\Phi\alpha = y$ has a solution $\hat{\alpha}$).

$\Phi \in \mathbb{R}^{N \times k}$, $y \in \mathbb{R}^N$, $(N > k)$

Given $\Phi \in \mathbb{R}^{N \times k}$, $y \in \mathbb{R}^N$

find $\hat{\alpha}$ s.t. $\Phi \hat{\alpha} = y$

i.e. solve the system of eq^{ns}

$\Phi \alpha = y$.

Range(Φ)

$$\begin{bmatrix} & \\ & \\ & \end{bmatrix} \alpha = \begin{bmatrix} \\ \\ \\ \end{bmatrix}$$

Φ $\quad$ y